DNS FROM THE FIRST PRINCIPLES

ALEXY KHRABROV

FIRST PAIR PRESS

DNS from First Principles

Names, Packets, Authority, Recursion, and the Design of rgbdns

Alexy Khrabrov

2026-07-30

Contents

Preface

DNS is often introduced as "the Internet's phone book." That metaphor is useful for about a minute. A phone book is one database, published in editions, mapping people to telephone numbers. DNS is a distributed protocol for finding typed, time-limited statements in a delegated tree. It has many writers, many readers, caches between them, multiple transports, and rules for proving both presence and absence. Names can point to addresses, but they can also identify mail exchangers, service endpoints, authoritative servers, cryptographic keys, and arbitrary text.

This book develops DNS from those underlying problems. The first half builds a mental model independent of any implementation. The second half walks through rgbdns, a memory-safe Rust reimplementation of the djbdns suite. The aim is not only to explain what each program does, but why its boundaries look the way they do: immutable compiled data for authority, a separate recursive cache, small diagnostic clients, foreground daemons, and stream-oriented logging.

The code is the final authority for rgbdns behavior. This book describes version 0.1.0 as built on 2026-07-23.

1 The problem DNS solves

1.1 Identity is not location

A network delivers packets to addresses. Humans and applications want stable identities. Those two things should not be fused.

Suppose a service is reached at `192.0.2.8`. If that address is embedded in every configuration, moving the service requires changing every client. A name such as `api.example` introduces indirection:

```
application → api.example → 192.0.2.8 → packets
```

Indirection has a cost: another system must answer the middle question. Its benefit is that the service owner can change the answer without changing the application. DNS is the globally deployed mechanism for this indirection.

The mapping is not a function from one name to one address. One name may have several addresses. The answers may differ by client location. A mail domain may name several mail exchangers with preferences. A service may delegate a subtree to another organization. The useful abstraction is therefore:

```
(owner name, record type, class) → a set of resource records
```

The owner and type together select an RRset. "RRset" means all resource records with the same owner, type, and class. Implementations should normally treat the set as a unit because caches and DNSSEC signatures do.

1.2 Requirements that pull in different directions

A global naming system must satisfy conflicting demands:

- It must scale without one central database receiving every query.

- Different organizations must control different parts of the namespace.
- Changes must propagate, but cached answers are essential for performance.
- Replies should usually fit in one datagram, but some answers are large.
- Old implementations must coexist with protocol extensions.
- A client needs to distinguish "no such name" from "that name has no record of this type."
- Operators need a way to transfer complete zones and to diagnose individual exchanges.

DNS answers these demands with hierarchy, delegation, caching lifetimes, compact binary messages, UDP plus TCP, explicit result codes, and typed records. Many operational surprises are direct consequences of those design choices rather than random quirks.

1.3 Roles, not just "DNS servers"

The phrase "DNS server" hides several jobs.

An **authoritative server** publishes data for zones it controls. It answers from configured facts and does not chase referrals on behalf of a client.

A **recursive resolver** accepts a question from a stub client, follows the delegation chain, validates and caches what it learns, and returns a final answer.

A **stub resolver** is the client-side library or program that sends a recursive query to a configured resolver.

A **forwarder** sends selected questions to another resolver rather than performing iteration itself.

Keeping these roles distinct is both conceptual hygiene and a security boundary. An authoritative daemon does not need a large mutable Internet-fed cache. A recursive resolver does not need the private machinery used to edit a zone. rgbdns follows the djbdns design and runs authority and recursion as different programs.

2 Names form a delegated tree

2.1 Labels and the root

A DNS name is a sequence of labels. In the presentation form `www.example.com.`, the dots separate the labels `www`, `example`, and `com`. The final dot represents the root's empty label. Reading from right to left walks from general to specific:

```
.                     root
└── com.               top-level domain
    └── example.com.   delegated domain
        └── www.example.com.
```

The absolute name has a wire limit of 255 octets, including length bytes and the terminating root label. Each ordinary label is at most 63 octets. DNS names are not inherently UTF-8 strings. Internationalized names are normally converted by applications into ASCII-compatible labels before DNS sees them.

DNS comparison is case-insensitive for ordinary ASCII letters, although the original spelling can be preserved. A robust implementation therefore needs a canonical comparison rule without losing the bytes required for encoding.

In rgbdns, `src/name.rs` represents a name as a vector of byte-vector labels. Construction validates label and total lengths. Parsing accepts the familiar dotted form and backslash escapes. The type provides parent and subdomain operations, case-insensitive ordering, display formatting, and wire encoding. Making invalid names difficult to construct removes repeated checks from the rest of the system.

2.2 Zones are administrative cuts

The namespace is one tree; a zone is an administratively served portion of that tree. The two are not identical.

The zone `example.com.` might contain records for `www.example.com.` and `mail.example.com.`, then delegate `research.example.com.` to other servers. The child remains below `example.com.` in the namespace but is outside the parent zone's authoritative contents.

A delegation is expressed by NS records at the cut. If a named server lies inside the delegated child, a resolver cannot first resolve that server's name through the child—it needs its address in order to reach the child. The parent therefore supplies an address record called **glue**. Glue is navigation data, not an assertion that the parent is authoritative for every fact about the host.

The root zone delegates top-level domains. A cold recursive resolver starts with a small configured set of root server addresses, asks the root where to find a top-level domain, asks that domain where to find the next child, and continues.

2.3 Wildcards are synthesis rules

A wildcard such as `*.example.com.` does not mean "return this record for every name ending in example.com." It participates only when the queried name does not exist, and the closest-encloser rules determine which wildcard, if any, can synthesize an answer. Existing intermediate names can block a wildcard.

rgbdns stores wildcard records under their literal wildcard owner and its zone lookup searches from the queried name toward the closest existing ancestor. It synthesizes the queried owner in returned records. This is more precise than a string suffix match and is one reason the `Zone` abstraction tracks known nodes in addition to records.

3 Resource records: typed facts with lifetimes

3.1 The common envelope

Every resource record has:

- an owner name;
- a numeric type;
- a class, almost always Internet class `IN`;
- a time to live, or TTL;

- type-specific data called RDATA.

The TTL is a lease offered to caches. If an authoritative server returns a TTL of 300 seconds, a cache may reuse that answer for at most five minutes before refreshing it. The TTL does not schedule a change and does not guarantee that every cache holds the answer for the full interval. It establishes an upper bound.

Changing a record and then lowering its TTL is too late for clients that already cached the older, longer lease. Planned migrations lower the TTL at least one old-TTL interval before the change, wait, make the change, and later raise it.

3.2 Core types

A maps an owner to an IPv4 address. **AAAA** maps it to an IPv6 address. Several records at one owner form an address RRset; DNS does not promise that clients use them in listed order.

NS names an authoritative server for a zone or delegation.

SOA, the start of authority, identifies the zone and carries operational parameters: primary server, responsible mailbox, serial, refresh, retry, expire, and negative-cache values. A secondary compares serial numbers to decide whether a transfer is needed. Serial arithmetic wraps in a defined 32-bit space, so blindly treating it as an ordinary integer can be wrong near the boundary.

CNAME says that its owner is an alias of another name. Except for DNSSEC and narrowly specified metadata, an owner with CNAME should not also hold unrelated data. A resolver follows the chain while defending against loops and excessive depth.

ANAME is not a standard DNS record type in rgbdns. It is authoritative server configuration that copies the address meaning of another name onto an owner. This distinction matters: clients receive ordinary A or AAAA records, not an ANAME record. ANAME can therefore provide address aliasing at a zone apex without violating CNAME exclusivity.

MX names a mail exchanger and gives it a preference. Lower numbers are preferred. The target is a name, not an address.

PTR provides a name-valued reverse mapping. IPv4 reverse names live below `in-addr.arpa.` with octets reversed. IPv6 reverse names live below `ip6.arpa.` with hexadecimal nibbles reversed.

TXT carries one or more length-delimited byte strings. Presentation formats often make it look like one free-form string, but the wire format retains segments.

SRV names a service endpoint with priority, weight, port, and target.

CAA constrains which certification authorities may issue certificates for a domain.

OPT is not ordinary zone data. It is a pseudo-record used by EDNS to negotiate UDP payload size and carry extension flags and options.

rgbdns models these forms with `RecordType`, `Record`, and the `RData` enum in `src/packet.rs`. Known structured types receive structured variants. Unknown types can remain opaque where

the format permits, preserving extensibility without confusing untrusted lengths with trusted objects.

3.3 Additional data is an optimization

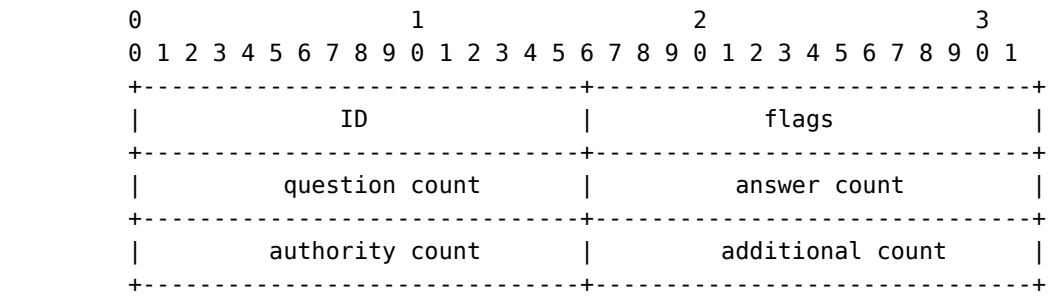
If an answer contains MX, NS, or SRV targets, the server may include associated A and AAAA records in the additional section. This can save queries. It does not change which RRset directly answers the question, and a resolver must apply the correct credibility rules rather than trusting unrelated additional data.

The rgbdns authoritative response path collects target names from those record types and adds locally available addresses. It de-duplicates targets before lookup and preserves the distinction between answers and helpful additionals.

4 Messages on the wire

4.1 The twelve-byte header

A DNS message begins with a fixed twelve-byte header:



The transaction ID lets a client associate a response with a query. Important flags include QR (query versus response), opcode, AA (authoritative answer), TC (truncated), RD (recursion desired), RA (recursion available), and the four-bit response code.

The four following sections contain questions, answers, authority records, and additional records. A normal question carries a name, requested type, and class. Resource-record sections add TTL, RDATA length, and RDATA.

All multibyte integers are network byte order. Every count and length comes from an untrusted peer. A decoder must prove that bytes exist before reading them, cap allocations, reject invalid labels and pointers, and finish with a coherent message rather than a partially trusted structure.

4.2 Name compression

Repeating full names would waste scarce datagram space. DNS permits a name suffix to be replaced by a two-byte pointer whose high bits are `11` and whose remaining bits are an offset earlier in the message.

Compression turns name decoding into graph traversal. A malicious packet can contain a pointer loop, excessive indirection, or an offset outside the packet. A safe decoder tracks visited offsets

or imposes a strict jump bound, checks every target, and separately enforces the expanded 255-octet name limit.

rgbdns's `Reader` in `src/packet.rs` keeps all reads within a borrowed byte slice. Name decoding validates pointer targets and bounds traversal. Record decoding confines each RDATA parser to the declared RDLENGTH. EDNS option iteration likewise checks the option header and value before advancing.

The `Writer` performs the reverse operation. Encoding is fallible: counts must fit 16 bits, names and RDATA must fit their fields, and the result must remain valid. This symmetry—decode into typed data, manipulate typed data, encode with checks—is the packet layer's central safety property.

4.3 Errors are protocol results

Several results that sound similar are materially different:

- **NOERROR with answers**: the requested RRset exists.
- **NOERROR without answers**, often called NODATA: the name exists but the requested type does not.
- **NXDOMAIN**: the queried name does not exist.
- **SERVFAIL**: the server could not safely complete processing.
- **REFUSED**: policy forbids the operation.
- **FORMERR**: the message is malformed.
- **NOTIMP**: the requested opcode is unsupported.

Negative answers normally include the zone's SOA so resolvers can cache the negative result. Confusing NODATA with NXDOMAIN can suppress other valid types at the same name.

rgbdns expresses authoritative lookup outcomes as `Lookup::Answer`, `Referral`, `NoData`, `NxDomain`, and `Refused`. That internal sum type forces the response builder to handle each protocol meaning explicitly.

5 UDP, TCP, EDNS, and truncation

5.1 Why there are two transports

Classic DNS uses UDP for ordinary queries because one request and one response need no connection setup. Traditional UDP DNS assumed a 512-byte message. Larger answers set TC, telling the client to retry over TCP. TCP frames every DNS message with a two-byte length.

Zone transfers use TCP. Modern responses—especially DNSSEC responses—often need more than 512 bytes, so EDNS lets a client advertise a larger UDP receive size through an OPT pseudo-record. Internet paths can still drop fragmented UDP packets. A commonly conservative payload is 1232 bytes, large enough for useful DNSSEC answers while fitting the IPv6 minimum MTU without fragmentation under normal headers.

TCP is not merely an emergency protocol. Firewalls that assume DNS is UDP-only break standards-compliant resolution.

5.2 Truncation must preserve a valid message

Cutting the last bytes off an encoded message creates a malformed packet. Correct truncation removes complete records, sets TC, updates section counts, and re-encodes. A useful removal order discards nonessential additional data before authority and answer data. OPT sometimes needs special treatment because it carries the EDNS response.

`src/server.rs` calculates a response limit from the caller's transport limit and the client's EDNS advertisement. It caps advertised UDP size, rejects multiple OPT records, responds to unsupported EDNS versions, and constructs a full typed response. If encoding exceeds the limit, `truncate` sets TC and removes complete records in a defined order until the packet fits. The same core response logic serves UDP and TCP without treating TCP as a giant UDP datagram.

6 Authority: answering from owned facts

6.1 Finding the closest relevant authority

For a question `(name, type)`, an authoritative server determines:

1. whether the name lies in a served zone;
2. whether a delegation cut is closer than that zone's apex;
3. whether the exact name exists;
4. whether the requested RRset exists;
5. whether a CNAME or wildcard changes the answer;
6. which SOA proves a negative result.

A query beneath a delegated child should produce a referral, not an authoritative negative answer from the parent. A query outside all configured zones should normally be refused. These boundary checks matter more than a simple map lookup.

rgbdns's `Zone` stores records in a `BTreeMap<Name, Vec<Record>>`, authoritative apices and delegation owners in ordered sets, and separate metadata for location and activation. Lookup walks name ancestry, recognizes cuts, filters visible records, applies exact-name and wildcard rules, and returns the typed `Lookup` outcome.

The response builder then:

- copies the query ID and relevant RD bit;
- marks authoritative answers with AA;
- expands CNAME chains with a 16-hop limit and visited-name set;
- adds address records for NS, MX, and SRV targets;
- clears AA on referrals;
- adds the SOA to negative answers;
- maps malformed, unsupported, and policy cases to protocol response codes.

The finite CNAME bound and visited set are deliberate denial-of-service and correctness controls. A cyclic zone must not turn one datagram into unbounded work.

6.2 tinydns data as a source language

djbdns uses a compact line-oriented zone source called `data`. The first character selects a record form. Common forms include:

Prefix	Meaning
`.`	zone authority plus NS and address data
`&`	delegation NS and optional glue
`Z`	explicit SOA
`=`	A plus matching reverse PTR
`+`	A only
`6`	AAAA plus reverse PTR forms
`3`	AAAA only
`@`	MX and optional exchanger address
`C`	CNAME
`A`	private ANAME (flattened A/AAAA alias)
`^`	PTR
`'`	TXT
`S`	SRV
`:`	generic record
`%`	client-location mapping

Fields are colon-separated with octal escapes for bytes that would otherwise be ambiguous. Optional fields carry TTL, timestamp, and location information. The format is terse because it was designed for mechanical generation as well as hand editing.

`Zone::parse` reads this language line by line. It ignores blank, comment, and disabled lines; reports the failing line number; expands convenience forms into ordinary typed records; validates IPv4, flat 32-digit IPv6, names, numeric ranges, and escaped bytes; and records authoritative and delegation structure. When an SOA serial is omitted, file loading derives a nonzero default from the source modification time.

Timestamp fields use TAI64-style cutoffs. Depending on the marker, a record can be visible before or after a specified instant. Location codes select records using configured client IPv4 prefixes. rgbdns carries that metadata beside the record and evaluates it at lookup time.

6.3 ANAME and apex address flattening

A zone apex necessarily owns SOA and NS data. A CNAME owner, by contrast, cannot also own ordinary records. A literal apex CNAME would therefore make the zone internally contradictory:

the alias rule says the owner has no other data while the authority rules require other data at that same owner.

Hosted sites still need a way for `example.com` to track addresses controlled by a platform such as `customer.blog-host.example`. DNS providers commonly call the solution CNAME flattening, ALIAS, or ANAME. The authoritative server resolves the configured target itself and publishes the resulting addresses under the configured owner.

rgbdns calls this feature **ANAME** and uses the private `A` source marker:

```
# Authority and nameserver address.
.example.com:192.0.2.53:ns1.example.com

# ANAME owner:target:maximum-ttl
Aexample.com:customer.blog-host.example:300

# Other apex data remains independent.
@example.com::mail.example.com:10:3600
'example.com:v=spf1 -all:3600
```

The form is:

```
Aowner:target:maximum-ttl
```

The TTL field is optional and defaults to 300 seconds. It is a ceiling, not a promise to extend the target's lifetime. If the upstream address has 45 seconds remaining and the ANAME limit is 300, rgbdns returns 45. If the upstream has 900 seconds remaining, rgbdns returns no more than 300.

ANAME is stored separately from ordinary `Record` values. It can coexist with SOA, NS, MX, TXT, CAA, and other non-address data. Zone validation rejects:

- A, AAAA, or CNAME data at the same owner;
- a wildcard ANAME owner;
- an owner that targets itself;
- different ANAME targets at one owner;
- a zero TTL.

The server only applies ANAME to A and AAAA questions. SOA, NS, MX, TXT, CAA, and all other questions continue through normal authoritative lookup. ANAME also does not override a delegation cut: a name beneath a delegated child still produces a referral from the parent.

For an address question, the response path is:

1. establish that ordinary authoritative lookup reaches the ANAME owner and does not cross a delegation;
2. query the configured recursive resolver for the target and requested address family;
3. validate response identity and framing;
4. follow only a connected CNAME chain beginning at the configured target;
5. collect the terminal A or AAAA RRset;
6. replace each terminal owner with the ANAME owner;

7. cap the remaining TTL and return an authoritative answer.

For example, an upstream result such as:

```
customer.blog-host.example. 180 IN CNAME edge.host.example.
edge.host.example.          120 IN A     192.0.2.80
```

becomes:

```
example.com.                120 IN A     192.0.2.80
```

The CNAME is deliberately absent. Consumers see a conventional authoritative address RRset at the apex.

The resolver cache is shared by requests handled by one server process. Positive entries expire with the upstream chain's shortest relevant TTL. Negative results use the authority SOA's negative TTL when available and 60 seconds otherwise. The configured ANAME ceiling is applied when constructing each response, so two owners may safely share a target while using different TTL policies.

Resolution is bounded in the same spirit as the rest of rgbdns:

- CNAME chains stop after 16 links;
- visited names detect cycles;
- no more than 64 terminal addresses are accepted;
- conflicting CNAME targets are rejected;
- upstream SERVFAIL and other resolver errors become authoritative SERVFAIL, not false NODATA;
- A and AAAA are cached independently.

`DNSCACHEIP` selects one or more recursive endpoints, separated by commas. Each endpoint may be an IP address using port 53 or an explicit socket address. Without it, rgbdns reads `/etc/resolv.conf`. A local validating `dnscache` is the preferred upstream when operators want DNSSEC validation and a cache shared with other local DNS work:

```
DNSCACHEIP=127.0.0.1:5354 IP=0.0.0.0 PORT=53 tinydns
```

ANAME metadata survives `tinydns-data` compilation through private CDB entries; it is not encoded as a made-up public RR type. This retains the source semantics across text and CDB operation without teaching ordinary DNS clients about a private wire format.

Standard AXFR has no interoperable way to describe this private policy. rgbdns therefore does not emit ANAME metadata in AXFR. An independently configured secondary needs the same ANAME source directive, while a conventional secondary can only serve address snapshots supplied through an external materialization workflow. Operators should account for that difference before treating ANAME zones as ordinary transferable zones.

6.4 CDB: compile once, read predictably

The traditional `tinydns-data` compiles text into a constant database, CDB. The serving process reads the compiled file instead of reparsing editable text for every startup or query. Compilation also enables atomic replacement: write and validate a new file, then rename it into place.

rgbdns's `src/cdb.rs` preserves the djbdns key/value layout for ordinary records and uses a private, NUL-prefixed key namespace for ANAME metadata. `compile` serializes typed records and metadata; `load` reads entries and reconstructs a `Zone`. The loader does not trust the database merely because it is local. It bounds file and entry sizes, validates keys, checks record layouts, decodes names and RDATA through explicit lengths, and rejects malformed data.

This is an important general rule: compiled configuration is still input. It may be truncated by a failed copy, generated by an older tool, or replaced by an attacker with filesystem access. Memory safety should not depend on the provenance story being perfect.

7 Recursion: discovering an answer

7.1 Iteration from the root

A recursive resolver turns one client request into a bounded sequence of queries. For `www.example.com. A`, a cold lookup is approximately:

```
stub → recursive resolver
          ├─ root:       who serves com?
          ├─ com server: who serves example.com?
          └─ example:    what is www.example.com A?
     ← final answer
```

The resolver follows referrals, resolves nameserver addresses when glue is insufficient, handles aliases, retries servers and transports, and detects loops. It caches useful RRsets so later clients may skip most of this path.

Root hints are not answers to every name. They are bootstrap addresses for reaching the root authority. They need periodic maintenance because the set can change, though names and anycast make changes infrequent.

7.2 The cache is part of correctness

A cache key includes at least name, type, and class. A cached positive RRset expires according to TTL. Negative results also have bounded lifetimes derived from SOA data. Delegation and nameserver-address caches help the iterative algorithm navigate efficiently.

Capacity is as important as time. An attacker can generate endless distinct names. An unbounded cache converts traffic into memory exhaustion. A practical resolver bounds response cache bytes, nameserver cache entries, recursion depth, referral work, packet sizes, concurrent operations, and timeouts.

`src/bin/dnscache.rs` uses Hickory's recursive zone handler inside rgbdns's process and policy shell. It configures:

- randomized query-name letter case;
- a bounded response cache, defaulting to 16 MiB;
- a bounded nameserver cache;
- bounded ordinary and nameserver recursion depth;
- a 1232-byte EDNS payload;
- UDP and TCP listeners;
- loopback-only clients by default, expanded through `ALLOW_NETS`.

Configuration values are parsed with explicit minimums and maximums. A typo such as an enormous cache size fails startup rather than silently allocating an operator's mistake.

7.3 Forward zones and djbdns roots

Private namespaces and split DNS often need selected suffixes sent to specific servers. rgbdns reads forward-zone configuration from the environment and the djbdns-style `ROOT/servers` directory. The filename identifies a suffix and the file lists bounded server addresses.

The special `servers/@` file represents root servers. Hickory consumes a root hints file in master-file syntax, so `PreparedRoots` translates djbdns's plain address list into a private temporary file. Creation uses restrictive permissions and cleanup occurs when the prepared object is dropped. This adapter preserves the external configuration contract without weakening the library boundary.

Forwarded private zones disable strict case-randomization response matching because legacy authorities may canonicalize owner case. They retain TCP retry and a bounded cache. This is a scoped compatibility decision, not a global removal of query hardening.

8 DNSSEC: authenticating the chain

8.1 What ordinary DNS cannot prove

Transaction IDs, source ports, and query-case randomization make blind spoofing harder, but they do not cryptographically establish who published an RRset. DNSSEC adds signatures and a chain of trust.

A zone signs RRsets with private keys and publishes DNSKEY records. A parent publishes a DS digest that identifies a child key. Starting from a configured root trust anchor, a validating resolver can authenticate the root DNSKEY, then a top-level domain's DS and DNSKEY, and so on to the answer.

RRSIG authenticates an RRset over a validity interval. DS links parent to child. NSEC or NSEC3 authenticates nonexistence by proving gaps in the ordered namespace. DNSSEC provides origin authentication and integrity; it does not encrypt queries or hide names.

Validation outcomes matter:

- **secure**: a valid chain reaches the answer;
- **insecure**: the chain proves that the child is unsigned;
- **bogus**: signatures or proofs fail;

- **indeterminate**: validation cannot be completed safely.

A resolver must not turn bogus data into a normal answer merely to improve availability. Clock correctness also becomes a dependency because signatures have inception and expiration times.

8.2 rgbdns validation policy

rgbdns configures the recursive handler with a static root trust anchor and DNSSEC validation enabled. Validation and NSEC3 work receive bounded caches and iteration policies. A failed validation surfaces as resolution failure rather than an unchecked answer.

The authoritative rgbdns data path focuses on the djbdns record surface; the recursive path is where DNSSEC validation is currently integrated. This is an example of honest component boundaries: "the suite supports validating recursion" does not imply that every authoritative signing workflow has been recreated.

9 Zone transfer and secondary service

9.1 AXFR is a stream, not a giant datagram

AXFR transfers a complete zone over TCP. A successful stream begins with the zone's SOA, contains the zone records, and ends with the SOA again. The records may span many DNS messages. A client must continue until it sees the closing SOA under the transfer rules; reading one response is insufficient.

Transfers reveal the zone contents and can consume resources, so authorities normally restrict clients. TSIG is a common authentication mechanism in the wider ecosystem, while IP allowlists are a simpler policy with weaker identity properties.

`src/axfr.rs` provides both sides. The standalone `axfrdns` command accepts TCP only and checks client networks, loopback by default. The packaged primary also routes AXFR through `tinydns`'s existing TCP listener when `ALLOW_NETS` is set. This is required when ordinary authoritative DNS and transfers must share one address on port 53: two separate processes cannot own that TCP endpoint.

Both entry points require one AXFR question, obtain a boundary-aware transfer from `Zone`, and frame bounded messages. `Zone::transfer` excludes records beneath delegated child zones and wraps the result in the apex SOA. The integrated listener applies its transfer allow-list only to AXFR; ordinary DNS-over-TCP remains reachable by all clients allowed through the network firewall.

`axfr-get` generates a random transaction ID, validates response identity and shape, collects records until the closing SOA, renders them in tinydns source form, writes a temporary output, and atomically installs the completed file. The temporary/final path pair prevents a failed transfer from replacing usable data with a partial zone.

10 The rgbdns program family

10.1 One suite, small purposes

rgbdns deliberately exposes separate commands:

Command family	Purpose
`tinydns`, `tinydns-data`, `tinydns-get`, `tinydns-edit`	authoritative service and data maintenance
`dnscache`	validating recursive resolver and cache
`axfrdns`, `axfr-get`	zone transfer server and client
`rbldns`, `rbldns-data`	address-prefix blocklist DNS
`pickdns`, `pickdns-data`	location-aware address selection
`walldns`	synthetic address/reverse answers
`dnsq`, `dnsqr`, `dnsip*`, `dnsname`, `dnsmx`, `dnstxt`	queries and diagnostics
`dnsfilter`, `dnstrace`, `random-ip`	stream lookup, delegation tracing, testing
`*-conf`	service-directory generation
`setuidgid`, `multilog`, `tai64n`, `tai64nlocal`	process and logging support

This composition makes privilege and failure boundaries visible. A compiler can run with write access to data while the server runs read-only. A recursive cache can be restarted without touching authority. Diagnostic clients reuse the packet and client libraries rather than embedding daemon behavior.

10.2 Specialized responders

`rbldns` treats the labels before a configured suffix as a numeric address, finds the most-specific matching IPv4 prefix in a compiled database, and returns configured A/TXT data. Parsing caps the number of numeric labels and validates networks before compilation.

`pickdns` maps client prefixes to two-byte locations and selects address sets for that location. It shuffles eligible addresses with operating-system randomness. Location-aware answers are a policy feature; clients behind shared resolvers may appear at the resolver's address, a limitation operators must understand.

`walldns` synthesizes narrowly defined forward and reverse answers without a zone database. These specialized services run through `src/special.rs`, which provides shared bounded UDP/TCP serving and passes the peer address to the handler.

The lesson is architectural: once parsing, transport, names, and record models are sound, unusual DNS policies can be small pure response functions rather than new monolithic servers.

11 Client behavior and diagnostics

11.1 A query is more than sending bytes

A DNS client creates a random transaction ID, encodes one question, sends it to an intended server, receives a response, and validates at least:

- source endpoint where the transport permits;
- transaction ID;
- QR and response shape;
- matching question;
- declared section lengths and names;
- truncation, with TCP retry when needed.

`src/client.rs` reads `DNSCACHEIP` or `/etc/resolv.conf`, supports IPv4 and IPv6 socket syntax, gets IDs from the operating system, applies UDP timeouts, rejects mismatched responses, and retries truncated UDP replies over TCP. The small command binaries format results for different use cases, while `dnsq` allows an explicit server and `dnsqr` uses recursive configuration.

`dnstrace` is conceptually different from a recursive lookup: it exposes the delegation path and intermediate authority so an operator can see where the chain stops. Good diagnosis asks four separate questions:

1. What did the stub send?
2. What did the recursive resolver cache or validate?
3. What delegation did the parent publish?
4. What does the authoritative server say directly?

Testing only the final application collapses all four layers and encourages guessing.

11.2 Practical checks

When a name fails, inspect type, server, flags, and authority section rather than asking only whether an address appeared.

```
dnsq A www.example.com 192.0.2.53
dnsq AAAA www.example.com 192.0.2.53
dnsq SOA example.com 192.0.2.53
dnstrace A www.example.com
```

Compare UDP and TCP when answers are large. Query the parent-side NS records and the child authority separately. An NXDOMAIN with an SOA is different from a timeout, SERVFAIL, or REFUSED, and each points to a different layer.

12 Security engineering in rgbdns

12.1 The packet is hostile

DNS combines nearly every parser hazard: nested lengths, compression pointers, variable counts, binary strings, recursive relationships, and network-facing availability requirements. “Written in

Rust" removes broad classes of memory corruption, but it does not automatically prevent allocation bombs, infinite loops, CPU amplification, path races, policy errors, or accepting incoherent messages.

rgbdns therefore uses several layers:

- `#![forbid(unsafe_code)]` for the library;
- explicit bounds before every wire read;
- validated `Name`, `Message`, and `RData` objects;
- limits on compression traversal, aliases, records, files, configuration lists, recursion, transfers, and cache sizes;
- cryptographic operating-system randomness for query IDs and selection;
- complete-record truncation;
- loopback-only defaults for recursion and transfer;
- atomic replacement for compiled databases and fetched zones;
- no shell interpolation when replacing a process.

Property tests in `tests/packet_properties.rs` feed arbitrary bytes to the decoder and exercise encode/decode invariants. Golden CDB fixtures compare compiled output with the expected djbdns layout. Network tests cross real UDP and TCP boundaries. Compatibility tests are valuable here because a parser can be safe yet subtly wrong, or compatible yet unsafe.

12.2 Least privilege and filesystem boundaries

The `*-conf` commands generate service directories whose run scripts execute the daemon under a selected account. rgbdns's `setuidgid` resolves the user and group, initializes supplementary groups, drops GID and UID, verifies the result, and directly replaces itself with the target program. Direct replacement preserves signals and exit status and avoids an extra shell-owned process.

Generated paths are shell-quoted and support binaries by absolute path. Configuration writers reject unsafe existing file types and apply intentional modes. CDB and AXFR update workflows install only complete outputs.

Privilege dropping is not a substitute for a restricted service account, read-only data, network policy, or supervisor hardening. It is one layer in a deployment.

13 Time and logs: TAI64N

13.1 Why a DNS suite contains time tools

Long-running daemons need logs, and djb's tools use TAI64N labels. A label has an `@`, sixteen hexadecimal digits of biased TAI seconds, and eight hexadecimal digits of nanoseconds:

```
@4000000037c219bf2ef02e94
```

TAI is a continuous atomic timescale. UTC inserts leap seconds, so converting between a POSIX/UTC timestamp and TAI requires the applicable TAI–UTC offset. The offset was 10 seconds at the Unix epoch convention used here and reached 37 seconds after the 2016 leap second.

`tai64n` timestamps each input line at the moment its first bytes are read. `tai64nlocal` recognizes a valid leading label and replaces it with local civil time at nanosecond precision. Invalid prefixes pass through unchanged. `multilog t` uses the same label generator, ensuring standalone filters and log files agree.

`src/tai64.rs` contains the complete 1972–2017 positive-leap transition table. It validates the fixed-width hexadecimal form and nanosecond range. Both stream filters use bounded memory: the localizer buffers only the 25-byte candidate prefix and streams the rest of even an extremely long line. At I/O failure the command exits 111, following the daemontools convention.

TAI64N provides sortable, unambiguous event labels. Converting to local time is a presentation step, not the archival representation.

14 Running rgbdns under supervision

14.1 The service contract

rgbdns daemons run in the foreground, emit diagnostics to standard error, take configuration from files and environment, and terminate on fatal startup errors. That is the portable contract a supervisor needs. The generated djbdns-style directories additionally provide `run` and `log/run` programs, but the daemon binaries do not require a particular supervisor.

The classic daemontools control plane is:

```
supervise service/       keep one process running
svc -u service/          bring it up
svc -d service/          bring it down
svc -t service/          send TERM
svstat service/          inspect status
```

No modern replacement is universally best. Choose according to the host and the migration boundary.

14.2 Recommendations

14.2.1 Existing Linux host: systemd

Use systemd when the machine already boots and manages services with systemd. It supplies dependency ordering, restart policy, socket and readiness models, resource controls, credential and filesystem sandboxing, a unified journal, and distribution-native administration. Avoid wrapping an rgbdns daemon in a second nested supervisor.

A minimal authoritative unit is:

```
[Unit]
Description=rgbdns authoritative DNS
After=network-online.target
Wants=network-online.target

[Service]
Type=simple
```

```
User=dns
Group=dns
Environment=IP=192.0.2.53
Environment=PORT=53
Environment=DATA=/etc/rgbdns/data.cdb
ExecStart=/usr/local/bin/tinydns
Restart=on-failure
RestartSec=1s
NoNewPrivileges=yes
ProtectSystem=strict
ProtectHome=yes
PrivateTmp=yes
ReadOnlyPaths=/etc/rgbdns
AmbientCapabilities=CAP_NET_BIND_SERVICE
CapabilityBoundingSet=CAP_NET_BIND_SERVICE

[Install]
WantedBy=multi-user.target
```

Prefer a socket above 1024 or a narrowly bounded bind capability over running the daemon as root. Test hardening settings on the target distribution because name-service libraries and trust-anchor paths may require additional read-only access.

Command mapping:

daemontools	systemd
`svc -u service`	`systemctl start service`
`svc -d service`	`systemctl stop service`
`svc -t service`	`systemctl kill --signal=TERM service`
`svc -h service`	`systemctl kill --signal=HUP service`
`svstat service`	`systemctl status service`
`multilog` output	journal, or explicit file logging policy

14.2.2 Closest service-directory migration: runit

Use runit when you want the smallest conceptual migration from daemontools. It uses a service directory with a `run` program, keeps the supervised process in the foreground, has a companion log service, and exposes the compact `sv` control command. Existing rgbdns generated `run` scripts are close to the required shape; adjust the directory layout and enablement symlink for the distribution.

daemontools	runit
`svc -u service`	`sv up service`
`svc -d service`	`sv down service`
`svc -t service`	`sv term service`
`svstat service`	`sv status service`

Choose runit for minimal hosts, appliances, or migrations where preserving the service-directory model matters more than rich dependency and sandbox policy.

14.2.3 Strong supervision composition: s6 and s6-rc

Use s6 when precise process supervision, reliable readiness, and composable small tools are primary requirements. Its `s6-supervise` and `s6-svc` are close in spirit to `supervise` and `svc`; `s6-rc` adds declared dependencies and transactional service-state changes. The ecosystem is particularly effective in carefully constructed containers and small systems, but its compilation and directory conventions make migration more involved than runit.

daemontools	s6
`supervise service`	`s6-supervise service`
`svc -u service`	`s6-svc -u service`
`svc -d service`	`s6-svc -d service`
`svc -t service`	`s6-svc -t service`
`svstat service`	`s6-svstat service`

Choose s6/s6-rc when the team is willing to own its service database and wants more rigorous dependency transitions than ad hoc shell orchestration.

14.2.4 OpenRC and container orchestrators

On an OpenRC-based distribution, use the native init integration unless there is a deliberate reason to introduce another supervision tree. OpenRC service scripts can use its supervisor support while retaining distribution-standard boot ordering and administration.

In Kubernetes or a similar orchestrator, run one foreground rgbdns daemon per container and let the platform own restart, health, resource limits, log collection, and rollout. Use a Deployment for `tinydns` or `dnscache`, a Service for stable network reachability, readiness/liveness probes that test the intended DNS role, ConfigMaps or mounted immutable CDBs for public data, and Secrets for sensitive material. Do not put systemd, daemontools, and the orchestrator around the same single process.

An s6-based container is reasonable only when one image intentionally contains several cooperating long-lived processes and that tradeoff is explicit.

14.3 A practical selection rule

Use this order:

1. Follow the host's native manager: systemd on systemd hosts, OpenRC on OpenRC hosts.
2. For a direct service-directory replacement, choose runit.
3. For a designed supervision graph or multi-process container, choose s6/s6-rc.
4. In an orchestrated single-process container, use the orchestrator.

The least risky migration preserves one owner for restart policy and logs. Running two supervisors creates ambiguous signal paths, duplicate restarts, and status commands that disagree.

15 Operating an authoritative service

15.1 Build, stage, verify, replace

A safe publication cycle separates source editing from serving:

```
cd /etc/rgbdns
tinydns-data
tinydns-get example.com A www.example.com
```

In production, compile in a staging directory, run representative exact, wildcard, delegation, negative, IPv4, IPv6, and large-response queries, then atomically replace `data.cdb`. Retain the previous known-good database for rollback. Query the bound service over both UDP and TCP after deployment.

For an ANAME zone, test the two address families and the unaffected apex record types separately:

```
dig @192.0.2.53 example.com A +norecurse
dig @192.0.2.53 example.com AAAA +norecurse
dig @192.0.2.53 example.com SOA +norecurse
dig @192.0.2.53 example.com MX +norecurse
```

The A and AAAA answers should have the apex as their owner and should not contain a CNAME. The SOA and MX answers should come entirely from zone data. Repeat the address queries after the target changes and after its TTL expires; this verifies refresh behavior rather than only the initial lookup. Also test the chosen recursive endpoint independently, because an authoritative ANAME lookup cannot succeed when its upstream resolver is unavailable.

Do not expose the recursive service to arbitrary networks by accident. The default `ALLOW_NETS` is loopback only because an open resolver can be abused for amplification and can consume local capacity. Likewise, expand AXFR allowlists only for intended secondaries.

15.2 Full deployment walkthrough: three authoritative topologies

This walkthrough installs one editable rgbdns primary and, when selected, one rgbdns secondary and/or BuddyNS. It deliberately shares one preparation, publication, and verification path among three useful topologies:

Topology	Published authorities	AXFR readers of `a`	AXFR readers of `b`
`a + BuddyNS`	`a` and the assigned BuddyNS names	BuddyNS	not applicable
`a + b + BuddyNS`	`a`, `b`, and BuddyNS	`b` and BuddyNS	BuddyNS, if configured as an alternate master
`a + b`	`a` and `b`	`b`	none

The common path is intentionally longer than any topology-specific branch. Choose the topology once, construct the corresponding NS and AXFR lists, then reuse the same installation and verification commands.

15.2.1 Names, addresses, and the security boundary

The examples use one service zone with in-bailiwick nameservers:

```
ZONE=example.net
PRIMARY_NS=a.ns.example.net
SECONDARY_NS=b.ns.example.net
PRIMARY_PUBLIC_IP=192.0.2.53
SECONDARY_PUBLIC_IP=198.51.100.53
PRIMARY_PRIVATE_IP=10.0.1.10
SECONDARY_PRIVATE_IP=10.0.2.10
```

Replace every documentation address and name. On AWS, bind each daemon to `0.0.0.0:53`; the guest normally sees its private interface while the Internet gateway maps its Elastic IP. Use private addresses for AXFR between instances in the same VPC. Give both instances stable public addresses before publishing delegation.

Permit public UDP 53 and public TCP 53 in the cloud security group and host firewall. Ordinary DNS needs both transports, so do not limit all TCP 53 to secondaries. rgbdns applies `ALLOW_NETS` only to AXFR questions. Separately allow TCP 53 from the secondary's private address or, preferably on AWS, its security group.

The examples place `a.ns.example.net` and `b.ns.example.net` inside the served zone, so the parent needs glue for both. Some deployments use names from a separate infrastructure zone. For example, `fieldnotes.es` can use `a.ns.cron.sh` and `b.ns.cron.sh`. In that case:

- put empty address fields on the `fieldnotes.es` NS lines;
- publish the `a.ns.cron.sh` and `b.ns.cron.sh` A records in the `cron.sh` zone, not in `fieldnotes.es`;
- create glue at the parent of `cron.sh` when those names are in-bailiwick nameservers for `cron.sh`; and
- remember that one packaged secondary instance synchronizes one zone.

Consequently, a `b` configured to transfer only `fieldnotes.es` must not be advertised as authoritative for `cron.sh`. Retain other working authorities for that infrastructure zone or create an additional explicitly managed secondary instance.

15.2.2 Obtain and install the packages

The GitHub Actions workflows publish architecture-specific artifacts. They are not APT or Zypper repositories. Install GitHub CLI and authenticate on a machine allowed to retrieve the artifacts:

```
gh auth login
```

Select the newest successful Debian build without depending on the version-specific `--status` option found only in newer GitHub CLI releases:

```
DEB_RUN_ID=$(
  gh run list -R querygraph/rgbdns \
    -w build-deb.yml -b master -L 50 \
    --json databaseId,conclusion \
    --jq '.[] | select(.conclusion == "success") | .databaseId' |
  head -n 1
)
mkdir -p "$HOME/rgbdns-deb"
gh run download "$DEB_RUN_ID" \
  -R querygraph/rgbdns \
  -n rgbdns-debian-amd64 \
  -D "$HOME/rgbdns-deb"
```

On the Debian or Ubuntu primary, install the downloaded package:

```
sudo apt install "$HOME"/rgbdns-deb/rgbdns_*_amd64.deb
sudo dpkg --audit
dpkg-query -W -f='${Status} ${Version}\n' rgbdns
```

The package intentionally replaces Debian’s djbdns and daemontools command packages because both suites own paths such as `/usr/bin/tinydns-get` and `/usr/bin/multilog`. Review APT’s removal plan before confirming on a host that already runs those services. Never use `dpkg --force-overwrite`.

For a selected topology containing `b`, download the newest successful openSUSE RPM artifact on the Leap 16 secondary:

```
RPM_RUN_ID=$(
  gh run list -R querygraph/rgbdns \
    -w build-rpm.yml -b master -L 50 \
    --json databaseId,conclusion \
    --jq '.[] | select(.conclusion == "success") | .databaseId' |
  head -n 1
)
mkdir -p "$HOME/rgbdns-rpm"
gh run download "$RPM_RUN_ID" \
  -R querygraph/rgbdns \
```

```
  -n rgbdns-opensuse-leap16-x86_64 \
  -D "$HOME/rgbdns-rpm"
RPM=$(find "$HOME/rgbdns-rpm/RPMS/x86_64" \
  -maxdepth 1 -name 'rgbdns-[0-9]*.x86_64.rpm' -print -quit)
rpm -K "$RPM"
sudo zypper --non-interactive --no-gpg-checks install "$RPM"
sudo rpm -V rgbdns
```

The artifact retains its `RPMS/x86_64` and `SRPMS` directories. Install the binary package under `RPMS/x86_64`; `SRPMS` contains the source RPM. The development package is payload-verified but not repository-signed, hence the explicit `--no-gpg-checks`. The RPM obsoletes and conflicts with RPM packages named `djbdns` and `daemontools`; inspect Zypper's transaction if either is installed.

Both packages create the non-login `rgbdns` user and group, protected configuration under `/etc/rgbdns`, state under `/var/lib/rgbdns/tinydns`, and hardened systemd units. Installation does not publish placeholder DNS data or enable authority.

Verify the installed account and units:

```
getent passwd rgbdns
getent group rgbdns
systemctl list-unit-files 'rgbdns-*'
```

15.2.3 Build one primary source file

Start with common application records and one SOA serial. A sortable `YYYYMMDDNN` serial is convenient; increment it before every publication. Construct the authoritative NS portion from exactly one topology block below.

Common records:

```
Zexample.net:a.ns.example.net:hostmaster.example.net:2026072901:16384:2048:1048576:2560:3600
+example.net:192.0.2.80:3600
C*.example.net:example.net:3600
```

For `a + BuddyNS`:

```
&example.net:192.0.2.53:a.ns.example.net:3600
&example.net::<BuddyNS name 1>:3600
&example.net::<BuddyNS name 2>:3600
&example.net::<BuddyNS name 3>:3600
```

For `a + b + BuddyNS`:

```
&example.net:192.0.2.53:a.ns.example.net:3600
&example.net:198.51.100.53:b.ns.example.net:3600
&example.net::<BuddyNS name 1>:3600
&example.net::<BuddyNS name 2>:3600
&example.net::<BuddyNS name 3>:3600
```

For `a + b`:

```
&example.net:192.0.2.53:a.ns.example.net:3600
&example.net:198.51.100.53:b.ns.example.net:3600
```

An & line with an address creates the NS record and its address/glue. The empty address fields on BuddyNS lines are intentional: those names belong to BuddyNS. Replace the account-specific placeholders with the exact names shown in BuddyBoard.

Store the assembled source as `/root/rgbdns.data` on the primary. Protect and compile a disposable copy before changing the service:

```
sudo install -o root -g root -m 0600 rgbdns.data /root/rgbdns.data
check_dir=$(mktemp -d)
sudo install -o "$(id -u)" -g "$(id -g)" -m 0600 \
  /root/rgbdns.data "$check_dir/data"
(cd "$check_dir" && tinydns-data)
ls -lh "$check_dir/data.cdb"
rm -r "$check_dir"
```

Compilation proves syntax and semantic consistency, not that the chosen addresses, delegation, mail policy, or application records are correct. Compare the new source with the old zone before cutover. AXFR from an existing authority is the best inventory when allowed; otherwise query all known record types and names from configuration management.

15.2.4 Construct the AXFR allow-list once

For a topology containing BuddyNS, copy BuddyBoard's current published transfer-source addresses into a protected, sourceable file. Express individual IPv4 sources as `/32` networks:

```
sudo install -o root -g root -m 0600 \
  buddyns-axfr.env /etc/rgbdns/buddyns-axfr.env
. /etc/rgbdns/buddyns-axfr.env
```

The file has this form:

```
BUDDYNS_AXFR_V4='203.0.113.10/32,203.0.113.11/32'
```

Treat the addresses as provider-maintained data, not constants copied forever from a book. Reconcile the file with BuddyBoard before a deployment and after provider network changes.

Choose the primary allow-list:

```
# a + BuddyNS
PRIMARY_ALLOW_NETS=$BUDDYNS_AXFR_V4

# a + b + BuddyNS
PRIMARY_ALLOW_NETS="$SECONDARY_PRIVATE_IP/32,$BUDDYNS_AXFR_V4"

# a + b
PRIMARY_ALLOW_NETS="$SECONDARY_PRIVATE_IP/32"
```

Run only the assignment for the selected topology. Never allow the entire VPC when one stable secondary address or security-group path is sufficient.

15.2.5 Configure and cut over the primary

First stage configuration without claiming port 53:

```
sudo rgbdns-setup primary \
  --data /root/rgbdns.data \
  --listen-ip 0.0.0.0 \
  --port 53 \
  --allow-nets "$PRIMARY_ALLOW_NETS" \
  --no-start
sudo -u rgbdns /usr/lib/rgbdns/compile-zone
sudo systemd-analyze verify \
  /lib/systemd/system/rgbdns-tinydns.service 2>/dev/null ||
sudo systemd-analyze verify \
  /usr/lib/systemd/system/rgbdns-tinydns.service
sudo ls -lah /var/lib/rgbdns/tinydns
sudo cat /etc/rgbdns/tinydns.env
```

The two unit paths cover Debian-family and openSUSE layouts. An unrelated legacy-unit warning from `systemd-analyze` does not invalidate a successful rgbdns unit check.

On a migration host, identify the existing owner of port 53:

```
sudo ss -lntup '( sport = :53 )'
```

Stop only the old authoritative services. Do not stop an entire `runsvdir` tree on a host where it also owns unrelated applications. For a classic djbdns layout:

```
sudo sv down /etc/axfrdns /etc/axfrdns/log
sudo sv down /etc/tinydns /etc/tinydns/log
sudo ss -lntup '( sport = :53 )'
```

Then enable rgbdns:

```
sudo systemctl enable --now rgbdns-tinydns.service
sudo systemctl status rgbdns-tinydns.service --no-pager --full
sudo ss -lntup '( sport = :53 )'
```

The authoritative daemon serves normal UDP, normal TCP, and allowed AXFR on the same port. Do not start the separately packaged `axfrdns` compatibility command on port 53.

Verify the primary locally and publicly before configuring delegation:

```
dig @127.0.0.1 example.net SOA +norecurse
dig @127.0.0.1 example.net NS +norecurse
dig @127.0.0.1 example.net A +norecurse
dig +tcp @127.0.0.1 example.net SOA +norecurse

dig @192.0.2.53 example.net SOA +norecurse
dig @192.0.2.53 example.net NS +norecurse
dig +tcp @192.0.2.53 example.net SOA +norecurse
```

Require `status: NOERROR`, the `aa` flag, the intended serial, the complete NS set, and correct address records.

15.2.6 Configure `b` when the topology includes it

The primary must allow the secondary's source address, and the network path must permit TCP 53, before this step. On the openSUSE secondary:

```
sudo ss -lntup '( sport = :53 )'
sudo rgbdns-setup secondary \
  --zone example.net \
  --primary 10.0.1.10 \
  --listen-ip 0.0.0.0
```

For `a + b + BuddyNS`, BuddyNS may read from `b` as an alternate master. In that case load the same current provider list and pass it to the secondary:

```
. /etc/rgbdns/buddyns-axfr.env
sudo rgbdns-setup secondary \
  --zone example.net \
  --primary 10.0.1.10 \
  --listen-ip 0.0.0.0 \
  --allow-nets "$BUDDYNS_AXFR_V4"
```

Choose one of those two setup commands. Setup performs a complete AXFR, validates the response and SOA bookends, atomically installs and compiles the zone, starts authority only after success, and enables a randomized five-minute refresh timer. It does not use NOTIFY or IXFR.

Check the one-shot synchronization result:

```
systemctl show rgbdns-secondary-sync.service \
  -p Result -p ExecMainStatus -p ActiveState -p SubState
```

A successful completed run reads:

```
Result=success
ExecMainStatus=0
ActiveState=inactive
SubState=dead
```

`inactive/dead` is correct for a finished `Type=oneshot` service. The `/run/rgbdns` runtime directory and its lock exist only while synchronization runs; systemd removes and recreates them for each invocation.

Verify service, timer, and answers:

```
sudo systemctl enable --now rgbdns-tinydns.service
sudo systemctl enable --now rgbdns-secondary-sync.timer
systemctl list-timers rgbdns-secondary-sync.timer
sudo ss -lntup '( sport = :53 )'
dig @127.0.0.1 example.net SOA +norecurse
dig @198.51.100.53 example.net SOA +norecurse
dig +tcp @198.51.100.53 example.net SOA +norecurse
```

The secondary serial must match the primary. To force a refresh after a publication:

```
sudo systemctl start rgbdns-secondary-sync.service
sudo journalctl -u rgbdns-secondary-sync.service -n 50 --no-pager
```

15.2.7 Configure BuddyNS when the topology includes it

In BuddyBoard:

1. add the zone;
2. configure `192.0.2.53:53` as a transfer master;
3. for `a + b + BuddyNS`, optionally add `198.51.100.53:53` as another master;
4. require the provider's transfer test to succeed; and
5. record the exact assigned BuddyNS names and transfer-source addresses.

The source zone's BuddyNS NS records, BuddyBoard's assigned names, and the eventual parent delegation must agree exactly. A provider transfer test should succeed before any registrar change.

From an allowed transfer source, or with a controlled temporary test address added to `ALLOW_NETS`, verify:

```
dig +tcp AXFR example.net @192.0.2.53
dig +tcp AXFR example.net @198.51.100.53  # when b permits BuddyNS
```

An unlisted client should receive `REFUSED`. Do not broaden the allow-list merely to make an arbitrary workstation AXFR test succeed.

15.2.8 Publish glue and delegation last

Create or verify registrar host objects before adding in-bailiwick nameservers:

```
a.ns.example.net = 192.0.2.53
b.ns.example.net = 198.51.100.53   # topologies containing b
```

Then publish the parent delegation matching the selected topology:

- `a + BuddyNS`: `a` plus the assigned BuddyNS names;
- `a + b + BuddyNS`: `a`, `b`, plus the assigned BuddyNS names;
- `a + b`: `a` and `b`.

Do not advertise `b` before it answers the current serial publicly. During a migration, keep old working secondaries in both the child NS RRset and parent delegation until new authorities pass UDP, TCP, SOA, and negative-answer tests. Remove old secondaries in a later serial change after parent updates have propagated.

Trace the parent and query every authority:

```
dig +trace +nodnssec example.net NS
dig @192.0.2.53 example.net SOA +norecurse
dig @198.51.100.53 example.net SOA +norecurse
```

Query each BuddyNS hostname as well when selected. Compare serials, NS RRsets, and authoritative flags. Also test a known name, a nonexistent name, UDP, and TCP:

```
dig @192.0.2.53 www.example.net A +norecurse
dig @192.0.2.53 does-not-exist.example.net A +norecurse
dig +tcp @192.0.2.53 www.example.net A +norecurse
```

15.2.9 Publish changes, upgrade, and recover

For each zone change:

1. edit the protected canonical source;
2. increment the affected SOA serial;
3. compile a disposable copy;
4. rerun `rgbdns-setup primary` with the complete allow-list;
5. query the primary;
6. force or await secondary refresh; and
7. compare every authority's serial.

Package upgrades preserve configuration and state. Upgrade a downloaded Debian package with `apt install /path/package.deb` and an RPM with:

```
sudo zypper --non-interactive --no-gpg-checks install \
  /path/to/rgbdns-VERSION-RELEASE.x86_64.rpm
sudo systemctl daemon-reload
sudo systemctl start rgbdns-secondary-sync.service
sudo rpm -V rgbdns
```

If secondary setup fails, it deliberately leaves authority stopped until the first valid transfer completes. Diagnose in this order:

```
sudo systemctl status rgbdns-secondary-sync.service --no-pager --full
sudo journalctl -u rgbdns-secondary-sync.service -n 100 --no-pager
dig +tcp @10.0.1.10 example.net SOA +norecurse
sudo cat /etc/rgbdns/secondary.env
sudo cat /etc/rgbdns/tinydns.env
```

A connection timeout points toward routes, security groups, or firewalls. `REFUSED` points toward `ALLOW_NETS` or an unexpected NAT source address. A validation error points toward the transferred zone. A successful one-shot with a refused local query means `rgbdns-tinydns.service` is not yet active.

After every reboot or upgrade, verify:

```
systemctl is-enabled rgbdns-tinydns.service
systemctl is-active rgbdns-tinydns.service
systemctl list-timers rgbdns-secondary-sync.timer
sudo ss -lntup '( sport = :53 )'
```

Monitor public UDP and TCP answers, unit restarts, timer failures, SOA convergence, transfer failures, and disk space. Keep the editable primary source in protected configuration management or backup. Secondary DNS improves serving availability; it is not a backup of the canonical source.

The distribution-specific deployment guides contain additional AWS, firewall, SELinux, and troubleshooting detail: `docs/DEBIAN.md` and `docs/OPENSUSE.md`.

15.3 Observe the right signals

Useful signals include:

- query and error rate by transport;
- truncated UDP responses and TCP retries;
- SERVFAIL, REFUSED, NXDOMAIN, and validation-failure rates;
- resolver cache capacity and latency percentiles;
- process restarts and file-descriptor use;
- root-hint and trust-anchor freshness;
- ANAME refresh latency, upstream failures, cache misses, and synthesized TTLs;
- time synchronization;
- CDB build identity and deployment time.

High NXDOMAIN volume is not automatically an incident; browsers, typo traffic, and discovery protocols generate it. A change from baseline paired with latency or SERVFAIL is more meaningful.

TAI64N log labels make events stable for storage. Convert them for human display at the edge:

```
tail -f main/current | tai64nlocal
```

16 Testing DNS software

16.1 Layers of evidence

Unit tests establish local invariants: name limits, record parsing, lookup outcomes, leap conversion. Property tests explore parser state spaces that examples miss. Golden fixtures establish compatibility with an external file format. Integration tests cross process and socket boundaries. Live interoperability tests compare behavior with independent clients and servers.

rgbdns uses all of these. A useful local sequence is:

```
cargo fmt --check
cargo test
cargo clippy --all-targets --all-features -- -D warnings
```

Network tests bind unprivileged loopback ports. CDB tests compile canonical fixtures and compare entries. Daemontools tests exercise process replacement, rotation, and TAI64 filter behavior. Packet properties assert that arbitrary input does not panic and that supported structured messages survive encode/decode round trips.

16.2 Adversarial cases worth keeping

Every DNS implementation should retain regression cases for:

- a compression pointer to itself or a pointer cycle;
- a pointer or RDATA length just beyond the packet;
- maximum-length labels and names;
- counts that cannot be satisfied by the remaining bytes;
- duplicate or malformed OPT records;

- tiny advertised transport limits;
- CNAME loops and excessive chains;
- ANAME self-reference, upstream CNAME loops, excessive address results, and resolver failure;
- wildcard names blocked by existing nodes;
- delegation cuts beneath an authoritative apex;
- NODATA versus NXDOMAIN;
- AXFR without a closing SOA;
- an enormous log line;
- configuration counts at and beyond each bound.

Tests should assert protocol meaning, not only that the process remains alive. A safe FORMERR is better than a crash, but a silent NOERROR can still be a serious bug.

16.3 Conformance as an executable specification

The conformance suite turns protocol prose into named, reviewable cases. Its scope is the DNS surface rgbdns implements; it does not imply support for every extension ever assigned by IANA. The principal coverage is:

Standard	Behavior exercised
RFC 1035	header identity, flags, names, compression, typed RDATA, UDP and TCP results
RFC 2181	in-bailiwick glue, coherent RRset TTLs, duplicate suppression, CNAME exclusivity
RFC 2308	NXDOMAIN versus NODATA, authoritative SOA, negative-cache TTL
RFC 3597	unknown QTYPE behavior and lossless opaque RDATA
RFC 4343	case-insensitive name identity with query-case preservation
RFC 4592	closest-encloser wildcard synthesis and empty non-terminals
RFC 5936	AXFR framing, identity, flags, SOA bookends, and zone boundaries
RFC 6891	one root-owned OPT, payload negotiation, DO, BADVERS, and unknown options
RFC 7766	TCP framing, connection reuse, pipelining, and full-size responses
RFC 8906	the authoritative-server matrix for unknown types, opcodes, flags, and EDNS fields
RFC 9619	exactly one question in a standard query

This is more useful than a single "RFC compliant" label. A test name identifies the rule, a packet fixture demonstrates it, and a failure points to a specific semantic regression.

RFC 8906 is especially valuable because it tests how a server behaves at the edges of what it understands. An unknown ordinary type is not a protocol error: the answer depends on whether the owner name exists. An unknown opcode is different. Because that opcode may define a body layout unlike QUERY, the server must produce NOTIMP from the header without first interpreting the body as an ordinary question. Unknown EDNS options are structurally validated and then ignored. An unsupported EDNS version produces BADVERS while retaining an OPT response.

The independent `drill` integration test supplies another boundary. It launches the real `tinydns` binary and asks the ldns client to make UDP, TCP, EDNS, mixed-case, and unknown-type queries. This catches accidental agreement between rgbdns's own encoder and decoder: the request and response cross an implementation boundary.

The complete focused matrix is:

```
cargo test --test rfc_conformance
cargo test --test wire_security
cargo test --test packet_properties
cargo test --test drill_interop
```

The generated suite exercises forty thousand cases per complete run. It feeds arbitrary bytes to the decoder, reparses every accepted packet, generates structured messages for semantic round trips, and changes ASCII letter case without changing DNS name identity. A separate truncation corpus tries every prefix of a valid structured packet. These properties do not prove the absence of all parser defects, but they explore combinations that hand-written examples rarely anticipate.

16.4 Hardening found by conformance work

Conformance testing improved the implementation rather than merely describing it.

The name decoder now records valid prior name boundaries. A compression pointer must be backward *and* must target one of those boundaries. Merely pointing at earlier bytes that happen to resemble a label sequence is rejected. This closes a class of ambiguous parses without forbidding legal compression.

Stub responses are bound to the request ID, QR bit, opcode, and exact question. TCP responses carrying TC are rejected. AXFR applies the same identity checks and additionally requires authoritative, non-truncated messages, controlled question repetition, an empty authority section, matching opening and closing SOAs, and records confined to the requested zone. These rules prevent a plausible-looking but unrelated response from being accepted as the answer to the outstanding operation.

Zone loading rejects a CNAME owner that also has other data and rejects multiple different CNAME targets. Before transmission, RRsets are normalized to their minimum TTL and duplicate records are removed. Negative answers cap the SOA TTL at the SOA MINIMUM field as RFC 2308 requires. EDNS OPT records in the wrong section and duplicate OPT records produce FORMERR.

The UDP and TCP daemons now share one bounded transport module. TCP connections carry deadlines, use a fixed worker pool, accept multiple framed queries, and support pipelined requests. This removes duplicated socket code while making RFC 7766 behavior an invariant shared by the authoritative and specialized servers.

16.5 Benchmarks and evidence-driven optimization

Correctness gates run before performance conclusions. The benchmark is a dependency-free stable-Rust harness in `benches/dns_core.rs`; the same harness is available as `examples/dns_core_bench.rs` for quick release-mode runs:

```
cargo bench --bench dns_core
RGBDNS_BENCH_ITERATIONS=10000 \
  cargo run --release --example dns_core_bench
```

It warms every operation, passes values through `std::hint::black_box`, and reports nanoseconds per operation. Measurements are comparable only on the same host, toolchain, power state,

and iteration count. Wire size is reported beside CPU time because DNS compression exchanges encoder work for fewer network bytes.

The July 2026 checkpoint used release mode on one aarch64 Android host:

Operation	Baseline	Optimized	Result
Encoded 64-record response	2,147 bytes	1,059 bytes	50.7% smaller
Decode small query	542 ns	458 ns	15.5% faster
Decode 64-record response	52,661 ns	29,540 ns	43.9% faster
Encode 64-record response	2,318 ns	5,309 ns	2.3 times slower
Exact lookup, 1,000 names	1,262 ns	1,244 ns	1.4% faster
NXDOMAIN, 1,000 names	29,889 ns	2,726 ns	11.0 times faster
Small authoritative response	17,007 ns	7,714 ns	54.6% faster
Truncate 200-record response	3,098,232 ns	2,570,077 ns	17.0% faster

Three structural changes explain most of the gains.

First, `Zone` maintains an index of every node, including empty non-terminals. A clearly absent name can return NXDOMAIN without scanning the records of a thousand-name zone. Conditional records still take the visibility path, so the index does not erase time or location semantics.

Second, response truncation searches the number of tail records to remove instead of encoding once for every removed record. It preserves the question and OPT record as long as possible and validates the final candidate against the transport limit.

Third, the packet writer records complete names and suffixes for RFC 1035 compression. RRsets tend to repeat the immediately preceding owner, so a last-owner cache avoids rebuilding and hashing suffix keys on the dominant path. The first compression design encoded the 64-record case in 34,075 ns; the cache reduced that to 5,309 ns.

The remaining encoder regression is intentional and visible. Compression makes the example packet roughly half as large while taking more local CPU than the old uncompressed writer. That is a defensible trade for an authoritative server because it reduces datagram pressure, TCP bytes, and downstream decode work. Recording the regression matters: optimization should reveal tradeoffs, not hide them behind one favorable number.

17 Reading the rgbdns source

17.1 A path through the code

Read the project in dependency order:

1. `src/name.rs` — the foundational name invariant.
2. `src/packet.rs` — types and bounded wire codec.
3. `src/zone.rs` — tinydns source and authoritative lookup semantics.
4. `src/cdb.rs` — compiled compatibility format.
5. `src/server.rs` — query validation, answer construction, transport limits.
6. `src/client.rs` — stub behavior and TCP fallback.
7. `src/axfr.rs` — streaming zone movement and atomic installation.
8. `src/dnscache_config.rs` and `src/bin/dnscache.rs` — iterative recursion, DNSSEC, forwarding, access, and resource policy.
9. `src/rbl.rs`, `src/pick.rs`, `src/wall.rs`, and `src/special.rs` — specialized responders.
10. `src/conf.rs`, `src/setuidgid.rs`, `src/multilog.rs`, and `src/tai64.rs` — deployment and operations.

The binaries in `src/bin` should then look thin. That is intentional. They parse the command contract, load configuration, call a library boundary, print diagnostics, and map fatal errors to the suite's exit convention.

17.2 Design patterns to carry elsewhere

Several rgbdns choices generalize beyond DNS.

Parse into valid types. If an invalid name can circulate as an ordinary string, every consumer must rediscover validation.

Bound dimensions independently. A packet byte limit does not replace a compression-depth limit; a cache byte limit does not replace a recursion-depth limit.

Separate policy from mechanism. `transport.rs` owns bounded UDP and TCP mechanics while the authoritative and specialized handlers own answer policy.

Compile mutable source into immutable serving data. This gives validation, atomic rollout, simple readers, and easy rollback.

Preserve protocol distinctions internally. A `Lookup` enum prevents NXDOMAIN, NODATA, referral, and refusal from collapsing into "no records."

Run in the foreground. It composes with old and new supervisors and keeps signals understandable.

Treat compatibility files as hostile. Historical layout fidelity need not mean historical trust assumptions.

Part II: Codebase exploration

The first part built DNS from its protocol obligations. This part reads rgbdns as a Rust system. Each chapter follows one execution path, names the abstraction that carries the obligation, and links directly to the implementation. The Obsidian edition adds live fragment cards beside these links: a reader can move from a claim to the exact symbol, then outward to the complete collocated source file.

18 Rust as a protocol-design tool

The important comparison with the original C is not that Rust uses newer syntax. It is that rgbdns can express DNS invariants at boundaries and have the compiler preserve them across the program.

The crate begins with `#![forbid(unsafe_code)]` in `src/lib.rs`. This is stronger than merely having no currently known unsafe block: a later contribution cannot introduce one without first making an explicit, reviewable change to the crate policy. The implementation still performs byte-level packet parsing, binary CDB loading, socket I/O, process credential changes, and concurrency. Those jobs do not require unchecked pointer arithmetic.

Rust improves the design along four axes.

Ownership makes lifetime and aliasing rules executable. A decoded `Message` owns its questions and records. A server borrows a `Zone` while constructing a response. Shared handlers use `Arc`, making cross-thread ownership visible in the type rather than implicit in process convention. There is no corresponding path for a response to retain a dangling pointer into the query buffer.

Algebraic data types preserve protocol distinctions. `RecordType` retains unknown numeric types as `Unknown(u16)`. `RData` distinguishes addresses, names, MX, SOA, TXT, SRV, CAA, EDNS, and opaque future data. `Lookup` distinguishes an answer, referral, NODATA, NXDOMAIN, and refusal. In C these states are commonly represented by related integers, nullable pointers, and out-parameters. In Rust, a `match` makes the cases locally exhaustive.

Fallible work is visible. Parsing and I/O return `Result`; optional facts return `Option`. The `?` operator propagates a typed failure without the unchecked sentinel values that make C error paths easy to omit. Conversion such as `u16::try_from(response.len())` documents the narrowing boundary and rejects overflow.

Concurrency composes with ownership. The bounded transport layer shares an immutable handler through `Arc<Handler>` and gives each TCP connection an exclusive mutable `TcpStream`. The compiler prevents a worker from retaining a borrow of a stack packet buffer after the buffer is reused.

Rust does not prove the DNS protocol correct. It removes broad classes of memory corruption and turns many design assumptions into compile-time or construction-time obligations. The remaining protocol work becomes visible enough to test directly.

19 Valid names instead of hopeful strings

`Name` is the first load-bearing abstraction. It stores a sequence of byte labels rather than a UTF-8 domain string. Its constructor is private to the module; all construction passes through parsing or `from_labels`, and both reach the same validation rule:

```
fn validate(labels: &[Vec<u8>]) -> Result<()> {
    if labels.iter().any(|l| l.is_empty() || l.len() > 63) {
        return Err(Error::InvalidName(
            "label must contain 1..=63 octets".into(),
        ));
    }
    let len = 1 + labels.iter().map(|l| l.len() + 1).sum::<usize>();
    if len > 255 {
        return Err(Error::InvalidName("wire name exceeds 255 octets".into()));
    }
    Ok(())
}
```

That small private function changes the rest of the codebase. `Zone` can use `Name` as a `BTreeMap` key without rechecking label lengths. The packet writer can calculate `wire_len` without wondering whether it will overflow the DNS name limit. `parent`, `suffix`, `wildcard`, and `is_subdomain_of` operate on labels rather than fragile dotted-string suffixes.

DNS identity is case-insensitive but responses should preserve the query's case. `Name` therefore retains original bytes while implementing `Eq`, `Hash`, and `Ord` with ASCII-folded comparisons. In C this invariant must be remembered by every hash table and comparison call site. Here it belongs to the key type.

This is a zero-surprise form of abstraction. It adds allocations when a name is built, but it removes repeated parsing and validation later. The benchmarked hot paths operate on already validated values, while the network boundary absorbs the cost once.

20 A bounded wire codec

DNS packets are attacker-controlled binary graphs: compression pointers can jump backward, names can share suffixes, section counts can lie, and RDATA lengths can disagree with actual bytes. `packet.rs` contains the codec and deliberately keeps the reader state small:

```
struct Reader<'a> {
    b: &'a [u8],
    p: usize,
    name_offsets: Vec<bool>,
}
```

The lifetime on `b` prevents the reader from outliving its input. Every scalar read uses slice bounds checks. Name decoding separately limits pointer hops, requires pointers to move backward, and records valid prior name boundaries. The last rule is stricter than merely checking that a pointer lands inside the packet: an interior byte can accidentally look like a valid label.

Decoded records become an `RData` variant. Unknown types are not discarded; they become opaque bytes paired with their numeric `RecordType`. This is the extension-safe behavior required by modern DNS. It also means an encoder can round-trip data it does not understand.

The writer uses compression, but optimization remains subordinate to a valid message. A last-owner cache handles repeated owners cheaply while suffix sharing reduces wire size across related names. The July 2026 benchmark shows the trade: a 64-record answer fell from 2,147 to 1,059 bytes, while compression made encoding slower than the uncompressed baseline. On DNS, fewer datagrams and less amplification surface can be worth several microseconds of local CPU.

Truncation uses a bounded search for the largest response that fits instead of repeatedly rebuilding one record at a time. The result preserves complete RRsets and required EDNS state. Performance work is therefore expressed as an algorithmic improvement behind the same `Message::encode` contract.

21 Zone data as an indexed semantic model

`Zone` is more than a parser for tinydns text. It is the semantic index used by authoritative answers:

```
pub struct Zone {
    records: BTreeMap<Name, Vec<Record>>,
    metadata: BTreeMap<Name, Vec<RecordMetadata>>,
    authoritative: BTreeSet<Name>,
    delegations: BTreeSet<Name>,
    locations: Vec<(Vec<u8>, [u8; 2])>,
    current_metadata: RecordMetadata,
    default_serial: u32,
    nodes: BTreeSet<Name>,
    unqualified_nodes: BTreeSet<Name>,
}
```

The maps hold records and djbdns location metadata. The sets encode facts that would otherwise require scans: zone apexes, delegation cuts, all existing nodes, and nodes that exist independently of location-qualified records. This all-node index is why the optimized NXDOMAIN benchmark is about eleven times faster than the earlier scan-based implementation.

The type also prevents semantic drift. Parsing validates CNAME exclusivity once. Lookup returns a `Lookup` enum, so absence cannot collapse into one null result:

- `Answer` carries the matching RRset.
- `Referral` carries delegation NS records and in-bailiwick glue.
- `NoData` says the name exists but the requested type does not.
- `NxDomain` says the name itself does not exist.
- `Refused` says the server is not authoritative for the question.

Wildcard processing uses the `nodes` index to find the closest encloser. Delegation processing uses the explicit `delegations` set. Transfer processing walks the same model while excluding child-zone contents. One representation therefore supplies ordinary answers, negative answers, wildcards, referrals, and AXFR without five subtly different interpretations of a zone.

22 From query bytes to an authoritative answer

`server::respond` is the central authoritative pipeline. Its shape is intentionally linear:

1. Reject an unknown opcode from the header without misparsing its body as a standard query.
2. Decode the packet, mapping malformed standard queries to `FORMERR`.
3. Enforce one question and valid OPT placement.
4. Derive the UDP response limit from EDNS and the transport ceiling.
5. Ask `Zone` for a typed `Lookup`.
6. Expand bounded CNAME chains and add relevant target addresses.
7. Normalize RRset TTLs and remove duplicates.
8. Encode or truncate the response.

The code separates mechanism from policy. `transport.rs` knows UDP datagrams, TCP length prefixes, timeouts, persistent connections, and a fixed worker bound. It knows nothing about zones. The handler knows DNS policy but receives transport limits and client identity as ordinary parameters. That separation lets specialized services reuse the network machinery without pretending to be authoritative zones.

The original djbdns family achieved robustness partly through small processes. rgbdns retains that decomposition while strengthening in-process boundaries. The binaries under `src/bin` are mostly adapters: environment, configuration, a library call, and the djbdns-compatible fatal exit convention. Small executables remain independently supervisable, but common logic is testable as ordinary Rust functions.

23 CDB compatibility without trusting the file

Compatibility is most valuable at the data boundary. rgbdns reads and writes the original tinydns `data.cdb` layout, so operators can preserve compilation and rollout habits. `cdb.rs` does not, however, inherit the old assumption that the compiled file is trustworthy.

The loader applies independent limits and checked arithmetic:

- the complete database is capped at one GiB;
- the 2,048-byte CDB header must exist;
- every hash-table position and slot count must fit inside the file;
- key and value lengths use `checked_add`;
- markers, locations, names, TTLs, cutoffs, and type-specific RDATA are validated before a `Record` enters a `Zone`.

This is a useful modernization pattern: preserve a durable external format, replace its implicit in-memory trust model. Operators gain compatibility; the serving process receives validated Rust values rather than pointers into a memory-mapped byte region.

Compilation likewise crosses an explicit boundary. A parsed `Zone` is written to a temporary CDB and then installed through the command workflow. Serving data is immutable between deployments. Rust's ownership does not itself make the rollout atomic, but it makes the stages—source text, validated model, compiled bytes, installed file—unambiguous.

24 Recursion by composition

Authoritative DNS is implemented in rgbdns's own small model. Recursive DNS, DNSSEC validation, caching, and upstream transport are composed from Hickory in `src/bin/dnscache.rs`. This is not a retreat from the rewrite; it is a deliberate abstraction boundary.

rgbdns owns policy that must remain djbdns-compatible or operator-visible: root hints, forwarding zones, allowed networks, cache budgets, recursion limits, EDNS payload, DNSSEC policy, listener addresses, and shutdown. Hickory supplies the complex iterative resolver machinery behind typed configuration and handler interfaces.

Every operator-controlled dimension is bounded. Cache sizes, recursion depth, name-server recursion depth, network lists, timeouts, and TCP message sizes have explicit limits. The `bounded_env` generic converts an environment value and verifies its range before server construction. A C implementation can do the same checks, but Rust makes the parsed type and the allowed range part of one reusable function.

Composition also improves performance engineering. The custom authoritative path stays small and directly benchmarkable. The resolver can use Tokio and a mature async DNS implementation without imposing that runtime on tinydns, rbldns, or walldns. Different concurrency models remain behind process and library boundaries rather than forcing one architecture across the suite.

25 Tests as executable protocol commentary

The strongest claims in this book have executable counterparts. `tests/rfc_conformance.rs` names normative requirements and constructs exact packets. `tests/wire_security.rs` contains a hostile corpus and checks every truncation of a structured packet. `tests/packet_properties.rs` generates arbitrary bytes and structured messages:

```
#[test]
fn arbitrary_packets_never_panic(
    bytes in prop::collection::vec(any::<u8>(), 0..4096)
) {
    let _ = Message::decode(&bytes);
}
```

The property suite does not prove that every accepted DNS packet has the desired meaning. It establishes three valuable invariants over a large input space: decoding never panics, accepted packets can be re-encoded and reparsed, and generated structured messages round-trip without semantic loss.

Golden CDB fixtures protect historical compatibility. Live UDP/TCP tests exercise connection reuse and framing. `drill` provides an independent encoder and decoder. The stable-Rust benchmark in `benches/dns_core.rs` measures the functions that rgbdns itself owns, and `docs/performance.md` records both timings and wire size.

This is where Rust most clearly changes the economics of a C rewrite. Memory safety removes many failure modes before testing. Property tests can then spend their budget on protocol

structure rather than rediscovering use-after- free and buffer-overrun variants. The remaining failures are more likely to be interesting DNS mistakes.

26 Abstractions and performance ledger

The rewrite's gains are not a single "Rust is faster" claim. Some changes buy safety, some buy clarity, and some measurably improve a hot path.

Design move	Rust expression	Operational effect
Valid names at construction	private fields, `FromStr`, `Result`	invalid labels cannot circulate
Complete DNS states	`RecordType`, `RData`, `Lookup` enums	unknown types survive; negative answers stay distinct
Bounded packet access	borrowed slices and checked indexing	malformed packets fail without memory corruption
Shared immutable service state	borrowing and `Arc`	explicit thread-safe ownership
Compatibility quarantine	checked CDB decoder into owned values	old files do not become trusted memory
Independent resource limits	typed bounded configuration	one limit cannot silently stand in for another
All-node zone index	`BTreeSet<Name>`	NXDOMAIN lookup improved about 11×
Compressed writer	bounded suffix reuse	repeated-owner answer became 50.7% smaller
Binary-search truncation	monotone fit search	200-record truncation improved 17%
Thin binaries	library functions plus small adapters	easier tests and independent supervision

The encoder example is especially important. Rust did not automatically make it faster: compression made encoding 2.3 times slower than the uncompressed baseline. But it halved the measured wire size, reduced fragmentation risk, and sharply improved the complete authoritative response path. Good systems engineering reports the trade rather than reducing it to a language slogan.

The deeper improvement over C is control. The data model says what is valid, the compiler checks ownership, the boundaries return typed failure, the tests state protocol properties, and benchmarks expose the remaining costs. That combination makes rgbdns easier to change without making DNS easier to fool.

27 Where DNS ends

DNS establishes named, cacheable facts and, with DNSSEC, their authenticated origin. It does not prove that the address belongs to the application a user intended, encrypt the subsequent connection, guarantee freshness inside the TTL window, or choose a healthy endpoint. TLS identity, application discovery, load balancing, routing, and monitoring build on DNS but remain separate systems.

That boundary is the best final replacement for the phone-book metaphor. DNS is a delegated publication and discovery protocol. Its tree assigns authority; its records carry typed statements; its TTLs make caching explicit; its packet format makes efficient exchange possible; recursion joins many authorities into one answer; DNSSEC authenticates the chain; and supervision keeps the implementing processes available without becoming part of the protocol.

rgbdns expresses those ideas as small programs over shared, validated Rust types. Understanding the protocol makes the program family unsurprising. Reading the program family, in turn, shows how the abstract DNS model becomes bounded packets, immutable databases, iterative queries, atomic files, and foreground processes.

28 Appendix A: Configuration quick reference

Common daemon variables include:

Variable	Meaning
`IP`	listen address
`PORT`	listen port
`DATA`	authoritative text or CDB path where supported
`ALLOW_NETS`	comma-separated client CIDRs for recursion or transfer
`DNSCACHEIP`	recursive endpoints used by client tools and ANAME flattening
`CACHESIZE`	bounded recursive response-cache capacity
`NSCACHESIZE`	bounded nameserver-cache entries
`RECURSION_LIMIT`	ordinary recursion depth
`NS_RECURSION_LIMIT`	nameserver-resolution recursion depth
`ROOT`	djbdns-compatible resolver configuration root

Use the command's `*-conf` generator as a starting point, then adapt the foreground `run` contract to the chosen native supervisor.

29 Appendix B: Further reading

The protocol is defined across many RFCs. A productive sequence is:

- RFC 1034, *Domain Names—Concepts and Facilities*.
- RFC 1035, *Domain Names—Implementation and Specification*.
- RFC 2181, clarifications including RRset and credibility rules.
- RFC 2308, negative caching.
- RFC 6891, Extension Mechanisms for DNS (EDNS(0)).
- RFC 7766, DNS over TCP requirements.
- RFC 4033, RFC 4034, and RFC 4035, DNSSEC.
- RFC 5155, NSEC3.
- RFC 5936, AXFR.
- RFC 1982, serial number arithmetic.

Implementation and operational references:

- The djbdns documentation: https://cr.yp.to/djbdns.html
- TAI64N format and tools: https://cr.yp.to/daemontools/tai64n.html
- s6 overview: https://skarnet.org/software/s6/overview.html
- s6-rc overview: https://skarnet.org/software/s6-rc/overview.html
- runit benefits: https://smarden.org/runit/benefits.html
- systemd project documentation: https://systemd.io/
- Hickory DNS: https://hickory-dns.org/